

Answer Key — Pressure: Force per Unit Area

Final answer with a one-line reason. Full methods are in the Notes' worked examples.

Objective

- Q1** (b) **force & divide; area** — pressure = force per unit area (F/A).
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- Q2** (b) **pascal** — pressure is measured in pascals ($1 \text{ Pa} = 1 \text{ N/m}^2$).
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- Q3** (b) **50 Pa** — $P = 200 \div 4 = 50 \text{ Pa}$.
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- Q4** (b) **reduce pressure by spreading force over a larger area** — larger area lowers the pressure on the shoulders.
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- Q5** (a) **Both true, R explains A** — the small sharp edge gives high pressure, so it cuts — exactly as R says.
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- Q6** **square** — $1 \text{ Pa} = 1 \text{ N/m}^2$ (newton per square metre).
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- Q7** **higher** — smaller area $\rightarrow$ higher pressure.
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Short answer

- Q8** Force acting per unit area ($P = F/A$); SI unit = pascal (Pa) = N/m^2 .
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- Q9** $P = 150 \div 5 = 30 \text{ Pa}$.
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- Q10** It spreads the weight over a larger area, reducing the pressure on the head and making it comfortable.
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Long / application

- Q11** (i) The pointed end has a tiny area, so it gives very high pressure for the applied force and pierces easily. (ii) Wide tyres spread the tractor's weight over a large area, lowering the pressure on soft soil so it doesn't sink.
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- Q12** On 6 m^2 : $P = 240 \div 6 = 40 \text{ Pa}$. On 2 m^2 : $P = 240 \div 2 = 120 \text{ Pa}$. The smaller face (2 m^2) exerts more pressure (120 Pa).
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